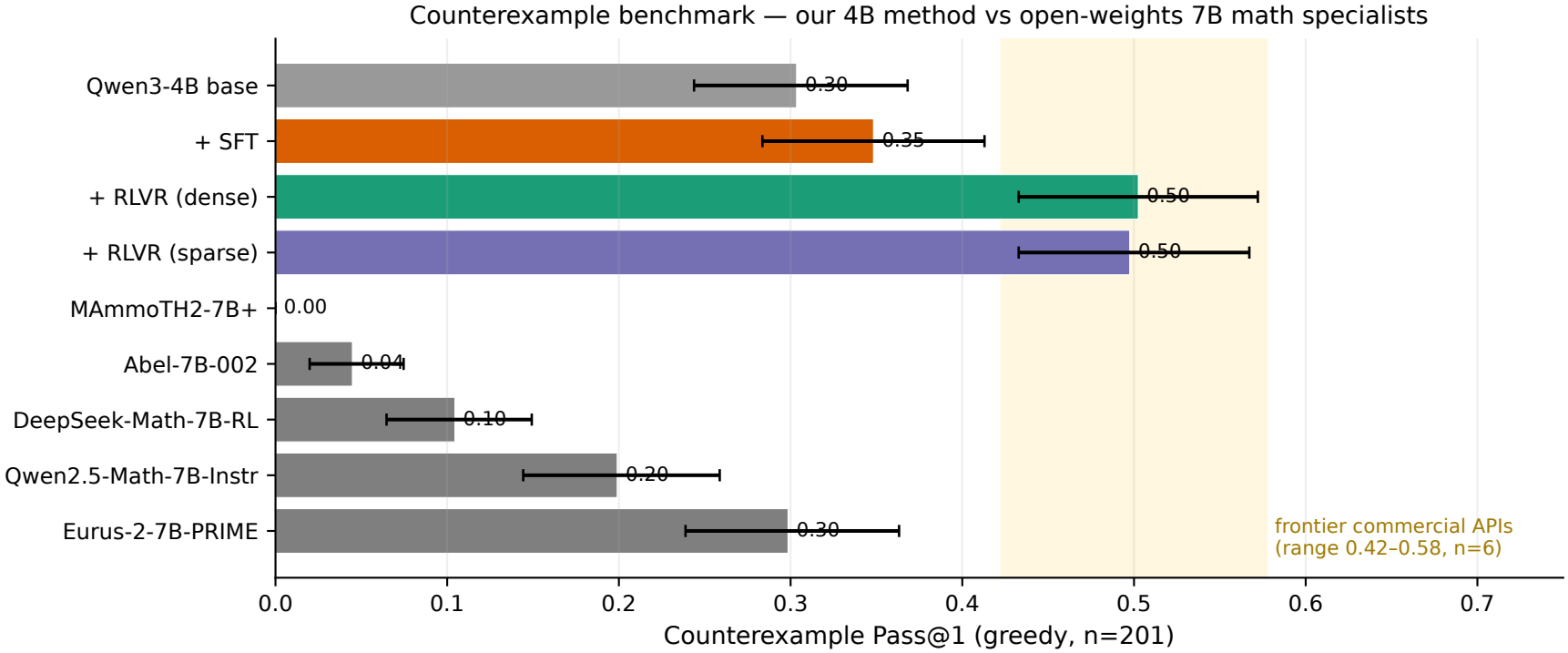
Counterexample benchmark — our 4B method vs open-weights 7B math specialists
Qwen3-4B base
+ SFT
+ RLVR (dense)
+ RLVR (sparse)
MAmmoTH2-7B+
Abel-7B-002
DeepSeek-Math-7B-RL
Qwen2.5-Math-7B-Instr
Eurus-2-7B-PRIME
0.30
0.35
0.50
0.50
0.00
0.04
0.10
0.20
0.30
frontier commercial APIs
(range 0.42–0.58, n=6)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Counterexample Pass@1 (greedy, n=201)